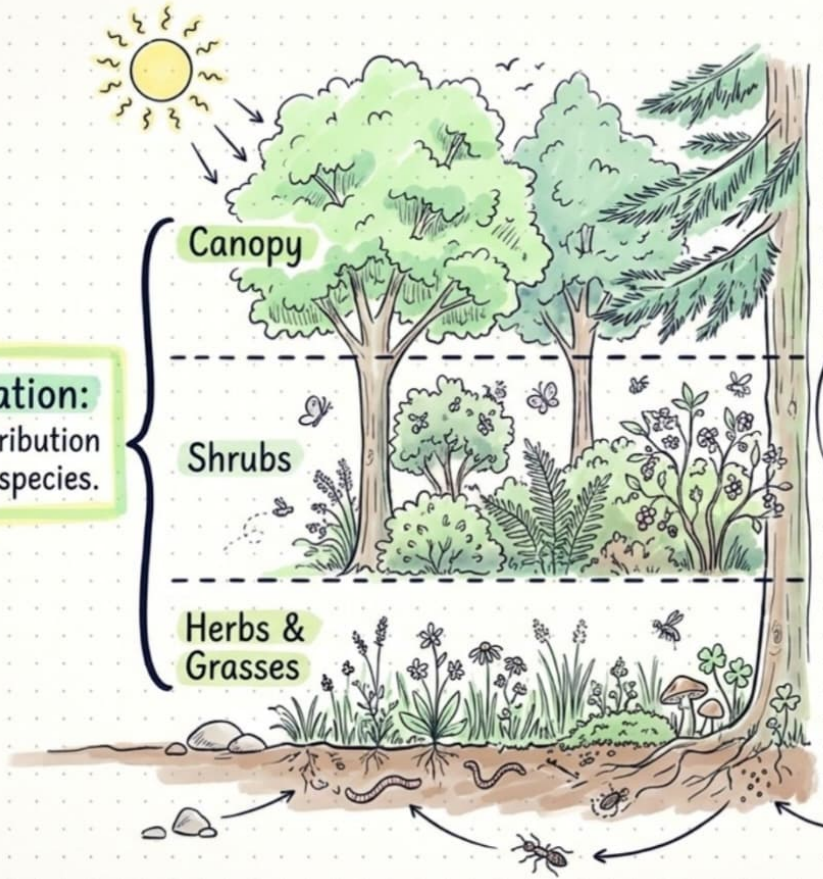
# Chapter 12: ECOSYSTEMS

## Structure & Function

Definition:
A functional unit of nature where living organisms interact among themselves and with their physical environment.

From small ponds to the global biosphere!

**Stratification:**
Vertical distribution of different species.




**Abiotic Components:**

Non-living physical factors (e.g., soil, water, temperature, sunlight).

**Biotic Components:**

Living components (e.g., plants, animals, microbes).

# Ecosystem Functions as a Unit

## 1. Productivity (Input)

## 2. Decomposition (Recycling)

## 3. Energy Flow (Transfer)

## 4. Nutrient Cycling

These 4 pillars operate together to keep the ecosystem self-sustaining.

# Pillar 1: Productivity (The Input)

**GPP** (Gross Primary Productivity) − **R** (Respiration Losses) = **NPP** (Net Primary Productivity)

**Secondary Productivity** = rate of new organic matter formed by consumers.

GPP = Total organic matter produced by photosynthesis.
NPP = Available biomass for heterotrophs (herbivores/decomposers).

Global NPP = ~170 Billion Tons.
Oceans occupy 70% of Earth, but only produce 55 Billion Tons!

# Case Study: The Pond Microcosm

**Autotrophs (Producers):** Phytoplankton & marginal plants.

**Abiotic:** Sun shining, dissolved nutrients in water, rich soil at bottom.

**Consumers:** Zooplankton & small free-swimming fish.

**Decomposers:** Fungi, bacteria, & flagellates in the mud.

**Insight:** The pond performs ALL ecosystem functions perfectly.
Inorganic → Organic → Consumed → Decomposed → Reused.

# Pillar 2: The 5 Steps of Decomposition

Raw Material = Detritus (dead leaves, animal remains, feces)

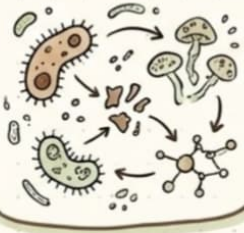
## Fragmentation

(Earthworm = Farmer's friend!)

## Leaching

(Water-soluble nutrients seep down)

## Catabolism



(Enzymes degrade detritus into simpler inorganic substances)

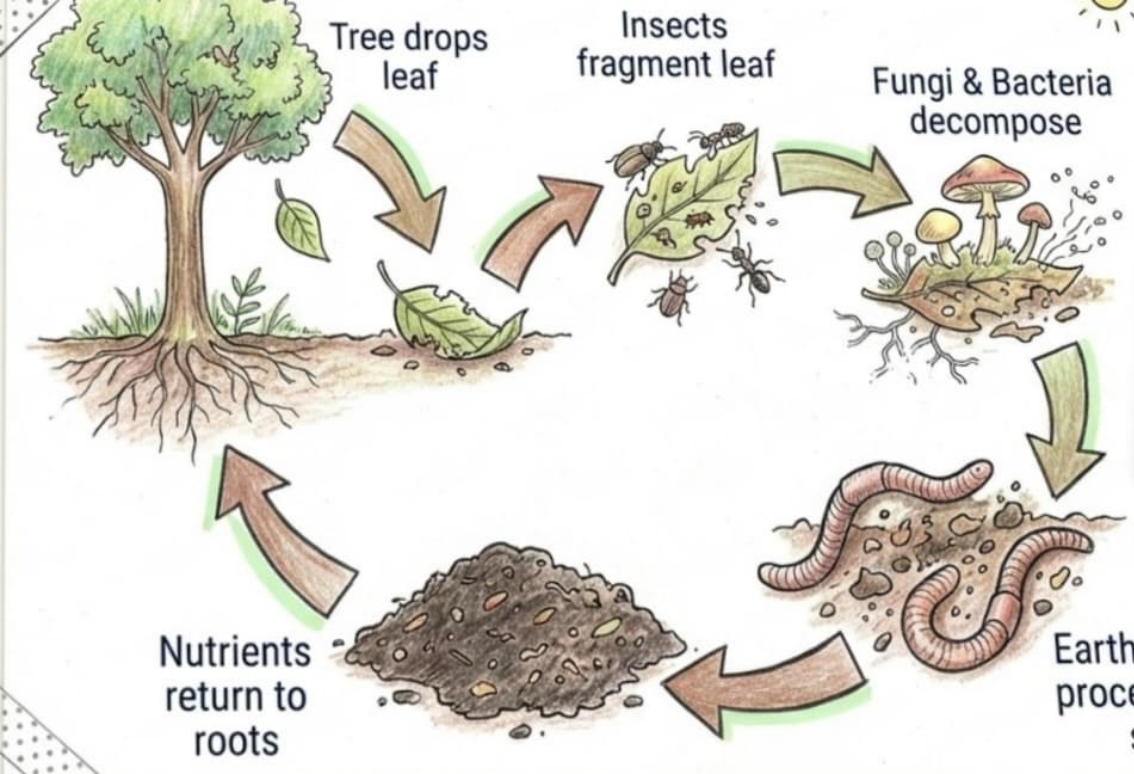
## Humification

(Accumulation of dark, resistant Humus. A nutrient reservoir)

## Mineralisation

(Microbes release final inorganic nutrients)

Note: These steps operate SIMULTANEOUSLY, not strictly one after another! Requires oxygen.




## What controls the speed?

- **Slower** if: rich in lignin & chitin, low temps, no oxygen.
- **Quicker** if: rich in nitrogen & sugars, warm & moist environments.

# Food Chains: GFC vs. DFC

| Grazing Food Chain (GFC) | Detritus Food Chain (DFC) |
| --- | --- |
| Starts with Producers (living plants). | Starts with Detritus (dead organic matter). |
| Relies on Herbivores & Carnivores. | Relies on Decomposers (Saprotrophs like fungi/bacteria). |
| Major energy conduit in Aquatic ecosystems. | Major energy conduit in Terrestrial ecosystems. |

In nature, they connect to form a **FOOD WEB**
(e.g., cockroaches & crows are omnivores connecting both).

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# Trophic Levels (Feeding Relationships)

**Standing Crop =** The mass of living material (biomass) at at each trophic level at a particular time.

**Step 1**

**Producers** (First Trophic Level) - Plants.

**Step 2**

**Primary Consumers** (Second Level) - Herbivores.

**Step 3**

**Secondary Consumers** (Third Level) - Carnivores.

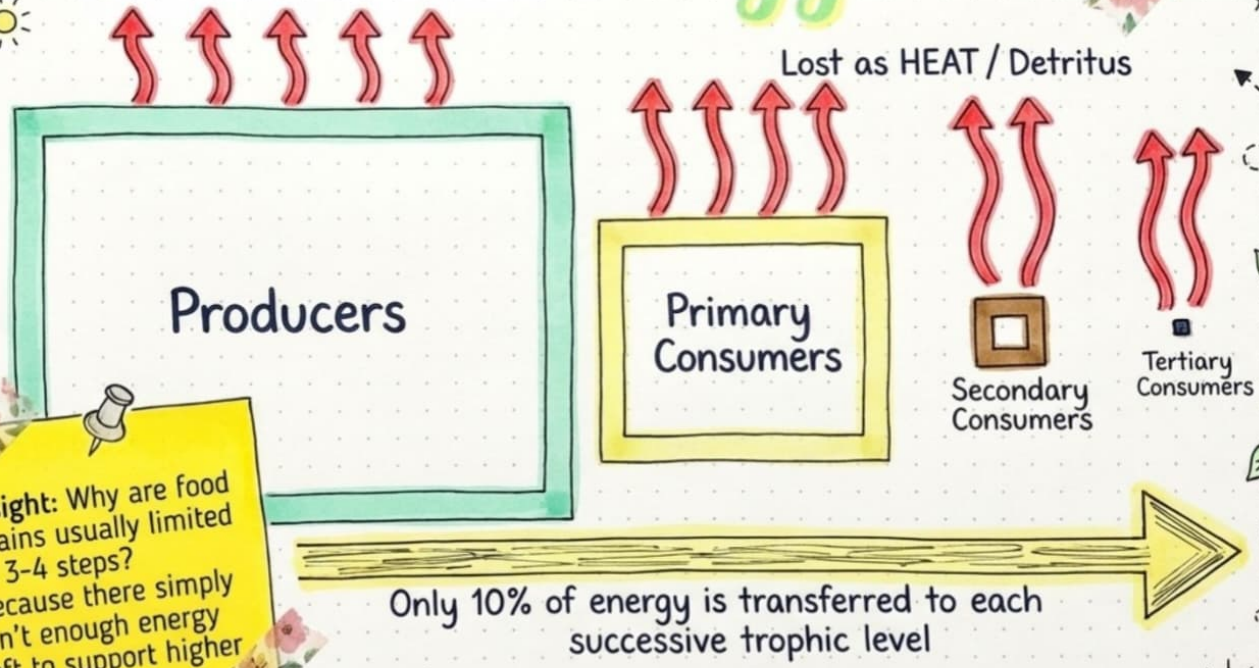
**Step 4**

**Tertiary Consumers** (Fourth Level) - Top Carnivores.

**Organisms can occupy multiple levels!**

Example: A sparrow eating seeds (Primary Consumer) vs. eating an insect (Secondary Consumer).

# The 10% Energy Law




**Insight:** Why are food chains usually limited to 3-4 steps? Because there simply isn't enough energy left to support higher levels!

# Pillar 4: Ecological Pyramids

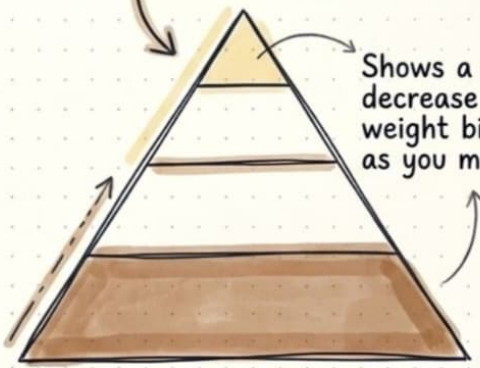
Base = Producers. Apex = Top Consumers.

## Pyramid of Numbers
(Grassland)

Energy flow shrinks!

Don't forget exceptions!

## Pyramid of Biomass
(Terrestrial)

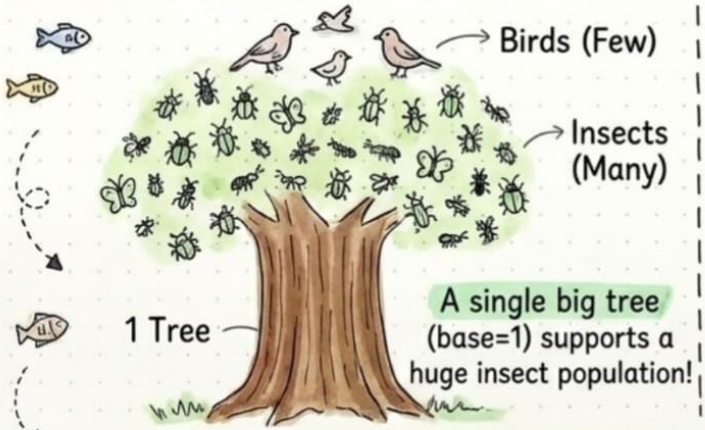


Shows a sharp decrease in dry weight biomass as you move up.

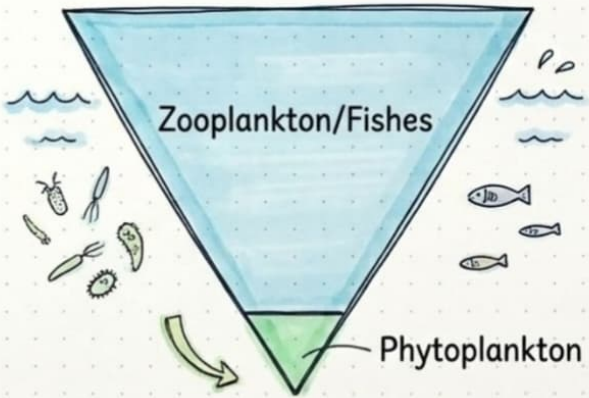
In most ecosystems, Number and Biomass pyramids are UPRIGHT (Base > Apex).

# The Exceptions: Inverted Pyramids

## Exception 1: Number Pyramid




## Exception 2: Biomass in Sea



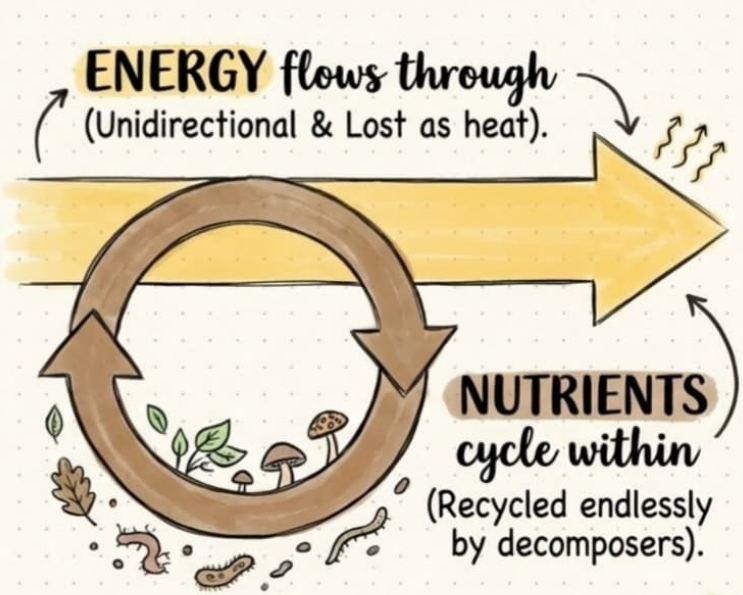

**The Paradox Explained:**

How does a small biomass of phytoplankton support heavy fishes?
Because phytoplankton reproduce incredibly fast! (Small standing crop, high turnover).

# Limitations & The Big Picture

## The Flaws of Pyramids:

1) They assume simple food chains (nature uses complex webs).
2) Same species can belong to multiple levels.
3) Saprophytes (decomposers) get ZERO place in pyramids despite being vital!

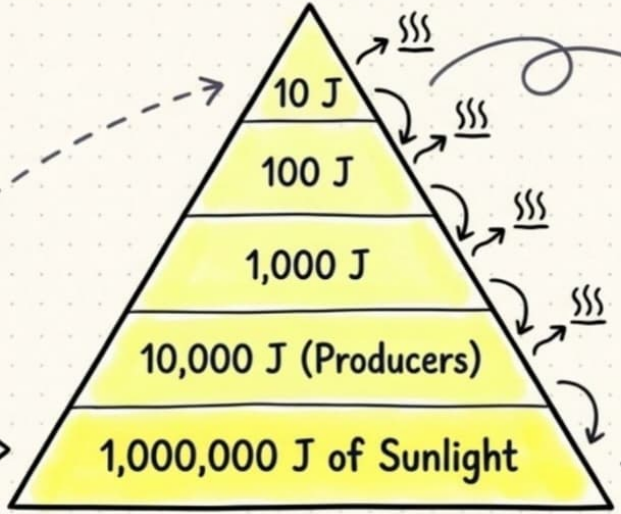



An ecosystem survives on constant solar energy input and endless terrestrial recycling.

NotebookLM

# Pyramid of Energy: The Absolute Rule

Primary producers convert only 1% of sunlight energy into NPP.




**Why?**

Because energy is always lost as heat at every single step (Thanks, 2nd Law of Thermodynamics!).

**The Pyramid of Energy is ALWAYS upright. It can NEVER be inverted.**